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Aryabhatta Maths Olympiad (AMO)

Level 1 — Practice Paper 2

CLASS 8

Questions	45 multiple-choice questions
Marks	100
Time	60 minutes
Section I	40 questions $\times$ 2 marks = 80 marks
Section II	5 Star Questions $\times$ 4 marks = 20 marks
Negative marking	No negative marking
Calculator	Not allowed
Chapters	Number System • Operations • Fractions, Decimals & Percentages • Algebra • Geometry • Mensuration • Data Handling • Applied Mathematics • Mathematical Reasoning

Instructions to candidates

- This paper has 45 multiple-choice questions. Each question has four answer choices, of which exactly one is correct.
- Section I (Q1–Q40) carries 2 marks per question. Section II (Q41–Q45) carries 4 marks per question.
- There is no negative marking. Unanswered questions score zero.
- Calculators are not permitted. Rough work may be done on a separate sheet.
- Mark your answers on the answer sheet at the end of the paper.
- All figures are not necessarily drawn to scale unless stated.

For parents and teachers

Print or view this paper in colour. Pictures are part of the questions.

Parts follow the AMO syllabus chapters. Chapter 7 is not part of the Class 8 syllabus, so the chapter numbers shown in the Part headings have gaps.

The answer sheet and the answer key are at the end. Remove only the answer key before the student begins.

Name:

School:

Roll No.:

SECTION I 40 questions × 2 marks each**Part 1 — Number System (Syllabus Chapter 1)****Q1.** Which of these numbers is rational?

- (A) $\sqrt{2} + \sqrt{2}$ (B) $\sqrt{0.9}$ (C) $(\sqrt{5})^3$ (D) $\sqrt{3} \times \sqrt{12}$

Q2. Simran spreads a square mat for a picnic in the park. The mat has an area of 2.89 m^2 . How long is each side of the mat?

- (A) 0.7225 m (B) 0.17 m (C) 17 m (D) 1.7 m

Q3. Which of these has the greatest value?

- (A) 0.9^2 (B) $\sqrt{0.64}$ (C) $\sqrt[3]{0.729}$ (D) $\sqrt[3]{0.343}$

Part 2 — Operations & Number Properties (Syllabus Chapter 2)**Q4.** Which of these numbers is negative?

- (A) $(-3)^{-2}$ (B) $(-2)^{-3}$ (C) 2^{-3} (D) $(-1)^{-4}$

Q5. $8^{-2} = 2^x$. What is the value of x ?

- (A) -6 (B) -4 (C) 6 (D) 1

Q6. A drop of water has a volume of $5 \times 10^{-2} \text{ mL}$. A water bottle holds $1.5 \times 10^3 \text{ mL}$. How many such drops are needed to fill the bottle?

- (A) 75 (B) 30,000 (C) 300,000 (D) 3,000

Part 3 — Fractions, Decimals & Percentages (Syllabus Chapter 3)**Q7.** Which of these deposits earns exactly ₹600 as simple interest?

- (A) ₹1,000 at 10% a year for 6 months (B) ₹4,000 at 6% a year for 2 years
(C) ₹2,000 at 7.5% a year for 4 years (D) ₹3,000 at 4% a year for 4 years

Q8. Jaya puts ₹8,000 in a bank that pays simple interest. After 9 months she gets ₹480 as interest. What is the rate of interest per year?

- (A) 6% (B) 4.5% (C) 12% (D) 8%

Q9. On sports day, 60% of the 400 students of a school run in races. Of the students who run, 25% win a medal. How many students win a medal?

- (A) 60 (B) 140 (C) 100 (D) 240

Q10. An item is marked ₹2,000. A shop can give two successive discounts in one of the four ways below. The second discount is on the reduced price. Which way makes the final price the lowest?

- (A) 25% and then 15% (B) 35% and then 5%
(C) 20% and then 20% (D) 30% and then 10%

Q11. The length of a rectangle is increased by 20% and its width is decreased by 10%. What happens to its area?

- (A) It increases by 8%. (B) It increases by 2%.
(C) It decreases by 12%. (D) It increases by 10%.

Part 4 — Algebra (Syllabus Chapter 4)

Q12. Which of these expressions is NOT a perfect square?

- (A) $9x^2 + 24x + 16$ (B) $x^2 + 10x + 25$
(C) $4x^2 - 12x + 9$ (D) $x^2 + 6x + 36$

Q13. For two numbers x and y , $x^2 - y^2 = 45$ and $x + y = 9$. What is the value of $x - y$?

- (A) 36 (B) 3 (C) 5 (D) 405

Q14. Mira cuts a ribbon 250 cm long into two pieces. One piece is 40 cm longer than twice the other piece. How long is the longer piece?

- (A) 180 cm (B) 145 cm (C) 105 cm (D) 70 cm

Q15. The expression $x^2 + kx + 24$ can be written as $(x + a)(x + b)$, where a and b are positive whole numbers. Which of these can NOT be the value of k ?

- (A) 14 (B) 12 (C) 11 (D) 10

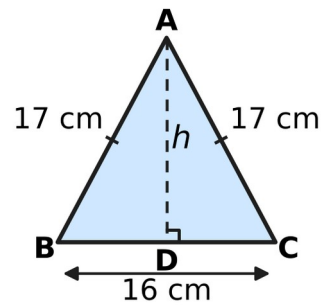
Q16. On a train journey, a coach had some passengers. At one station, $\frac{3}{5}$ of them got off and 12 new passengers got on. The coach then had 44 passengers. How many passengers were in the coach before the station?

- (A) 110 (B) 32 (C) 80 (D) 140

Part 5 — Geometry (Syllabus Chapter 5)

Q17. In the picture, $\triangle ABC$ has $AB = AC = 17$ cm and $BC = 16$ cm. AD is perpendicular to BC , so D is the midpoint of BC . What is the height $h = AD$?

- (A) 25 cm (B) 8 cm (C) 9 cm (D) 15 cm

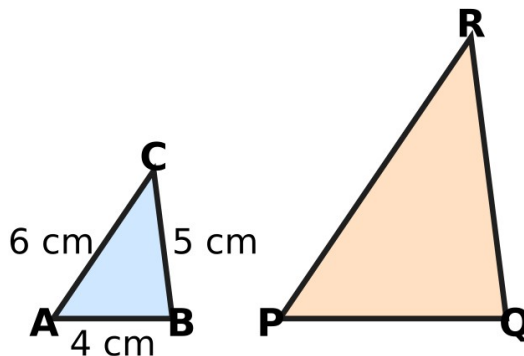


Q18. $ABCD$ is a parallelogram. Which of these facts is enough to prove that $ABCD$ is a rectangle?

- (A) One of its angles is 90° . (B) Its opposite angles are equal.
(C) Two of its adjacent sides are equal. (D) Its diagonals are perpendicular.

Q19. Which of these solids has exactly 30 edges?

- (A) a prism whose base has 8 sides (B) a prism whose base has 10 sides
(C) a pyramid whose base has 10 sides (D) a prism whose base has 15 sides

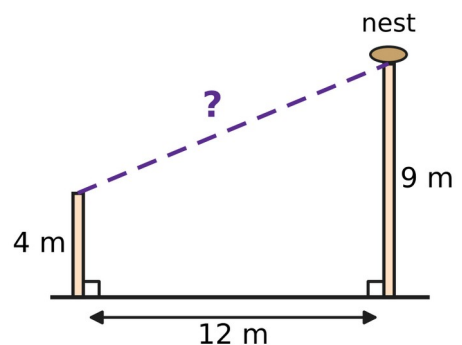


Q20. In the picture, $\triangle ABC \sim \triangle PQR$, with $AB = 4$ cm, $BC = 5$ cm and $CA = 6$ cm. The perimeter of $\triangle PQR$ is 45 cm. What is the length of the longest side of $\triangle PQR$?

- (A) 36 cm (B) 15 cm (C) 12 cm (D) 18 cm

Q21. Two upright poles stand on level ground 12 m apart. They are 4 m and 9 m tall, as in the picture. A bird flies in a straight line from the top of the shorter pole to its nest on top of the taller pole. How far does it fly?

- (A) 17 m (B) 15 m (C) 13 m (D) 5 m



Part 6 — Mensuration (Syllabus Chapter 6)

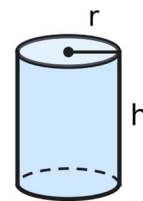
Cone	K	L	M	N
Radius (cm)	6	5	2	3
Height (cm)	2	3	15	8

Q22. The table gives the radius and the height of four solid cones. Which cone has the greatest volume?

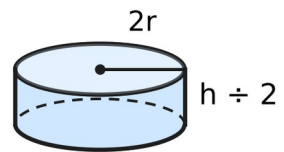
- (A) Cone K (B) Cone L (C) Cone M (D) Cone N

Q23. Cylinder A has radius r and height h . Cylinder B has radius $2r$ and height $h \div 2$, as in the picture. The volume of B is how many times the volume of A?

- (A) 4 (B) 8 (C) 2 (D) 1



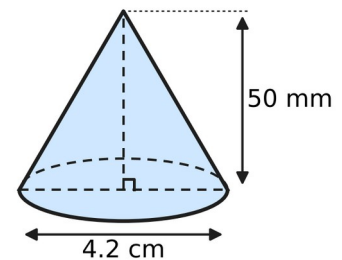
Cylinder A



Cylinder B

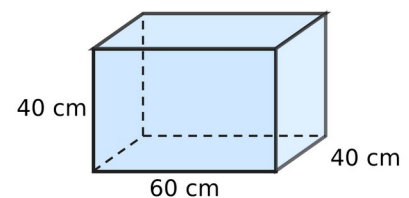
Q24. A solid cone of candle wax, shown in the picture, has a base diameter of 4.2 cm and a height of 50 mm. What is its volume? (Use $\pi = \frac{22}{7}$.)

- (A) 231 cm³ (B) 23.1 cm³ (C) 69.3 cm³ (D) 11 cm³



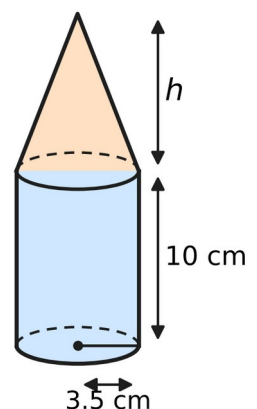
Q25. Lena fills an empty aquarium tank, a cuboid 60 cm × 40 cm × 40 cm as in the picture, using a jug that holds 1.2 litres. How many full jugs of water does she need to fill the tank? (1 litre = 1,000 cm³.)

- (A) 80 (B) 800 (C) 8 (D) 96



Q26. Anya buys a solid wooden toy rocket from a toy shop. It is a cylinder with a cone of the same radius on top, as in the picture. The radius is 3.5 cm and the cylinder is 10 cm tall. The whole rocket has a volume of 500.5 cm³. How tall is the cone, h ? (Use $\pi = \frac{22}{7}$.)

- (A) 19 cm (B) 39 cm (C) 3 cm (D) 9 cm

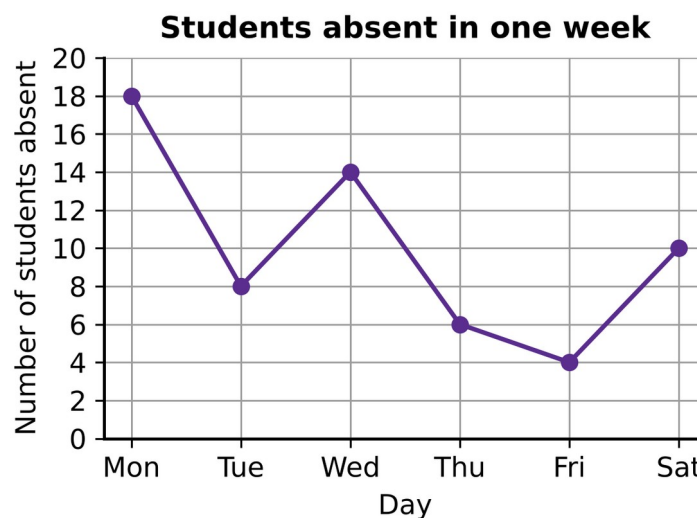


Part 7 — Data Handling, Statistics & Probability (Syllabus Chapter 8)

Favourite subject	Maths	Science	English	Art
Number of students	120	160	90	110

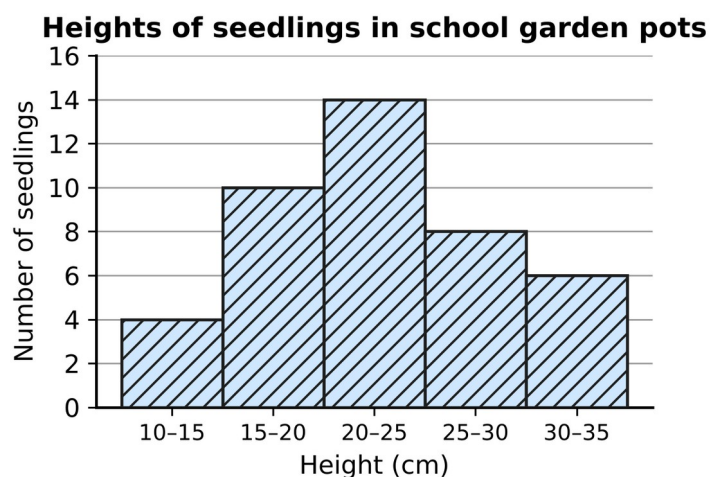
Q27. The table shows the favourite subjects of 480 students. The data is to be shown in a pie chart. Which subject will have a sector of 90° ?

- (A) Maths (B) Science (C) English (D) Art



Q28. The line graph shows the number of students absent from a school on each day of one week, Monday to Saturday. What is the mean number of students absent per day?

- (A) 11 (B) 10 (C) 9 (D) 12



Q29. The histogram shows the heights of the seedlings in the school garden pots. The class 10–15 means 10 cm or more but less than 15 cm. A frequency polygon is drawn for this histogram. Which of these points (height in cm, number of seedlings) lies on the frequency polygon?

- (A) (15, 10) (B) (17.5, 14)
(C) (17.5, 10) (D) (20, 10)

Part 8 — Applied Mathematics (Syllabus Chapter 9)

Item	A	B	C	D
Cost price (₹)	80	150	50	200
Selling price (₹)	100	180	65	236

Q30. The table shows the cost price and the selling price of four items. Which item gives the greatest profit percent?

- (A) Item A (B) Item B (C) Item C (D) Item D

Q31. Ashok borrows ₹36,000 at 8% per year simple interest. How much interest does he pay for each month?

- (A) ₹240 (B) ₹288 (C) ₹3,000 (D) ₹2,880

Q32. Rekha sells a table and makes a profit of ₹180. This profit is 15% of her cost price. What is her selling price?

- (A) ₹1,020 (B) ₹1,200 (C) ₹2,700 (D) ₹1,380

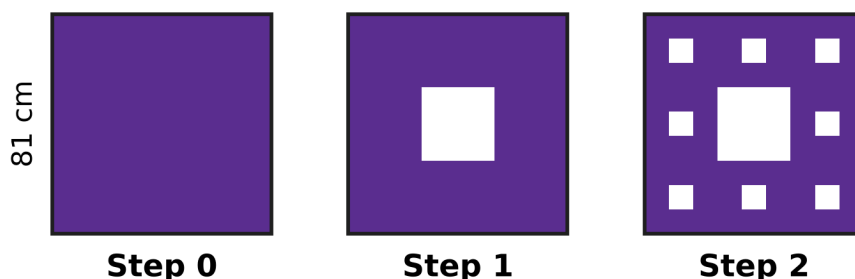
Q33. At the fruit market, Irfan buys 30 kg of oranges at ₹60 per kg and 20 kg of bigger oranges at ₹90 per kg. He mixes them and sells all 50 kg at ₹81 per kg. What is his profit percent?

- (A) 35% (B) 12.5% (C) 9% (D) 8%

Q34. Pavan sells a wall clock. His profit is one-fifth of the selling price. What is his profit percent on the cost price?

- (A) 125% (B) 80% (C) 20% (D) 25%

Part 9 — Mathematical Reasoning & Problem-Solving (Syllabus Chapter 10)



Q35. The picture shows the first steps of the Sierpiński carpet. It starts at Step 0 with a filled square of side 81 cm. At each step, every filled square is cut into 9 equal squares and the middle one is removed. What is the side of each smallest filled square at Step 3?

- (A) 3 cm (B) 1 cm (C) 9 cm (D) 27 cm

Q36. How many two-digit numbers have digits that differ by 7?

- (A) 4 (B) 6 (C) 5 (D) 3

Q37. Beads are threaded on a closed necklace in the colours red, blue, green, yellow, red, blue, green, yellow, ... all the way round. The pattern never breaks, even where the last bead meets the first. Which of these could be the number of beads?

- (A) 42 (B) 50 (C) 54 (D) 48

Q38. In the multiplication $ABCD \times 4 = DCBA$, each letter stands for a digit. Different letters stand for different digits, and neither A nor D is 0. What is the four-digit number ABCD?

- (A) 1,089 (B) 2,178 (C) 2,198 (D) 2,187

Q39. Tejas puts 10 pairs of shoes on a shoe rack with a top, a middle and a bottom shelf. Each shelf gets at least 1 pair and at most 5 pairs. In how many ways can he choose how many pairs go on each shelf?

- (A) 18 (B) 21 (C) 36 (D) 4

Q40. Which of these numbers is divisible by 36?

- (A) 6,354 (B) 5,736 (C) 5,832 (D) 5,382

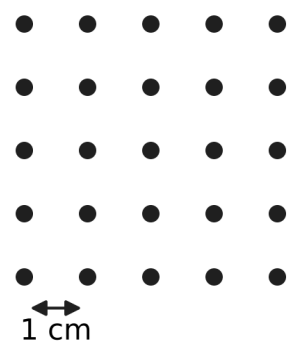
SECTION II — Star Questions 5 questions × 4 marks each

Q41. Take a two-digit number. Add its two digits, multiply its two digits, and add these two results and 12. For some numbers the answer is the number itself. For example, 23 works, because $(2 + 3) + (2 \times 3) + 12 = 23$. How many two-digit numbers work, 23 included? [4 marks]

- (A) 6 (B) 3 (C) 9 (D) 4

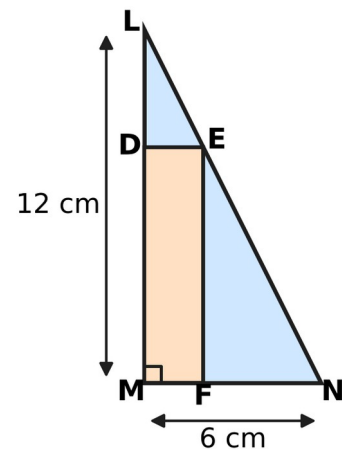
Q42. The picture shows 25 dots in a 5×5 square pattern, 1 cm apart. A square is drawn with all four corners on dots; its sides need not be along the rows of dots. How many different squares of area exactly 5 cm^2 can be drawn? [4 marks]

- (A) 16 (B) 0 (C) 8 (D) 4



Q43. In the picture, $\triangle LMN$ has a right angle at M, $LM = 12$ cm and $MN = 6$ cm. A rectangle MDEF is drawn with D on LM, F on MN and E on LN. The area of the rectangle is 16 cm^2 , and its sides MD and MF have different lengths. What is the perimeter of the rectangle? [4 marks]

- (A) 20 cm (B) 16 cm (C) 24 cm (D) 10 cm

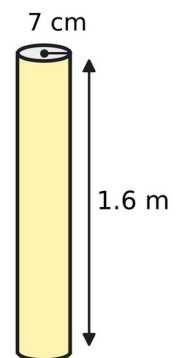


Q44. The list price of a classroom desk is ₹3,000. Vijay, a dealer, buys desks from the maker at successive discounts of 25% and 20% on the list price. He sells them to a school at 10% off the list price, and gives 1 desk free with every 4 desks the school pays for. What is Vijay's profit percent? [4 marks]

- (A) $16\frac{2}{3}\%$ (B) 20% (C) $33\frac{1}{3}\%$ (D) 50%

Q45. Gaurav paints the curved surfaces of 25 cylindrical fence posts. Each post has radius 7 cm and height 1.6 m, as in the picture. The owner pays him ₹50 for every square metre painted. The paint costs Gaurav ₹704 in all. What is his profit percent on the cost of the paint? (Use $\pi = \frac{22}{7}$.) [4 marks]

- (A) 80% (B) 20% (C) 25% (D) 150%



Answer Sheet AMO Level 1 • Class 8 • Practice Paper 2**Name:****School:****Roll No.:***Shade or tick one circle for each question.*

- | | | |
|---------------------|---------------------|---------------------|
| 1. (A) (B) (C) (D) | 16. (A) (B) (C) (D) | 31. (A) (B) (C) (D) |
| 2. (A) (B) (C) (D) | 17. (A) (B) (C) (D) | 32. (A) (B) (C) (D) |
| 3. (A) (B) (C) (D) | 18. (A) (B) (C) (D) | 33. (A) (B) (C) (D) |
| 4. (A) (B) (C) (D) | 19. (A) (B) (C) (D) | 34. (A) (B) (C) (D) |
| 5. (A) (B) (C) (D) | 20. (A) (B) (C) (D) | 35. (A) (B) (C) (D) |
| 6. (A) (B) (C) (D) | 21. (A) (B) (C) (D) | 36. (A) (B) (C) (D) |
| 7. (A) (B) (C) (D) | 22. (A) (B) (C) (D) | 37. (A) (B) (C) (D) |
| 8. (A) (B) (C) (D) | 23. (A) (B) (C) (D) | 38. (A) (B) (C) (D) |
| 9. (A) (B) (C) (D) | 24. (A) (B) (C) (D) | 39. (A) (B) (C) (D) |
| 10. (A) (B) (C) (D) | 25. (A) (B) (C) (D) | 40. (A) (B) (C) (D) |
| 11. (A) (B) (C) (D) | 26. (A) (B) (C) (D) | 41. (A) (B) (C) (D) |
| 12. (A) (B) (C) (D) | 27. (A) (B) (C) (D) | 42. (A) (B) (C) (D) |
| 13. (A) (B) (C) (D) | 28. (A) (B) (C) (D) | 43. (A) (B) (C) (D) |
| 14. (A) (B) (C) (D) | 29. (A) (B) (C) (D) | 44. (A) (B) (C) (D) |
| 15. (A) (B) (C) (D) | 30. (A) (B) (C) (D) | 45. (A) (B) (C) (D) |

Q41–Q45 are Section II Star Questions (4 marks each).

For the parent or teacher

Section I correct: _____ / 40 × 2 = _____ marks

Section II correct: _____ / 5 × 4 = _____ marks

Correct: _____ / 45 **Score:** _____ / 100

Answer Key and Solutions AMO Level 1 • Class 8 • Practice Paper 2**Quick answers**

1 D	6 B	11 A	16 C	21 C	26 D	31 A	36 C	41 D
2 D	7 C	12 D	17 D	22 B	27 A	32 D	37 D	42 C
3 C	8 D	13 C	18 A	23 C	28 B	33 B	38 B	43 A
4 B	9 A	14 A	19 B	24 B	29 C	34 D	39 A	44 B
5 A	10 B	15 B	20 D	25 A	30 C	35 A	40 C	45 C

Solutions

Q	Part • level	Answer	Why?
1	Part 1 • Ch1 • Easy	(D) $\sqrt{3} \times \sqrt{12}$	$\sqrt{3} \times \sqrt{12} = \sqrt{36} = 6$, which is rational. The others are $2\sqrt{2}$, $\sqrt{0.9}$ and $5\sqrt{5}$, which are irrational.
2	Part 1 • Ch1 • Easy	(D) 1.7 m	Side = $\sqrt{2.89} = 1.7$ m, because $1.7 \times 1.7 = 2.89$.
3	Part 1 • Ch1 • Medium	(C) $\sqrt[3]{0.729}$	$\sqrt{0.64} = 0.8$, $\sqrt[3]{0.729} = 0.9$, $0.9^2 = 0.81$ and $\sqrt[3]{0.343} = 0.7$. The greatest is $\sqrt[3]{0.729} = 0.9$.
4	Part 2 • Ch2 • Easy	(B) $(-2)^{-3}$	$(-2)^{-3} = 1/(-2)^3 = 1/(-8) = -1/8$, which is negative. The others are $1/8$, $1/9$ and 1.
5	Part 2 • Ch2 • Easy	(A) -6	Since $8 = 2^3$, $8^{-2} = (2^3)^{-2} = 2^{-6}$, because $3 \times (-2) = -6$. Check: $8^{-2} = 1/64 = 2^{-6}$.
6	Part 2 • Ch2 • Medium	(B) 30,000	Number of drops = $(1.5 \times 10^3) \div (5 \times 10^{-2}) = 0.3 \times 10^5 = 3 \times 10^4 = 30,000$.
7	Part 3 • Ch3 • Easy	(C) ₹2,000 at 7.5% a year for 4 years	SI = $P \times R \times T \div 100$: $2,000 \times 7.5 \times 4 \div 100 = ₹600$. The others earn ₹50, ₹480 and ₹480.
8	Part 3 • Ch3 • Easy	(D) 8%	Time = $9/12 = 3/4$ year. $R = 100 \times 480 \div (8,000 \times 3/4) = 48,000 \div 6,000 = 8\%$ per year.
9	Part 3 • Ch3 • Easy	(A) 60	Students who run = 60% of 400 = 240. Medal winners = 25% of 240 = 60.
10	Part 3 • Ch3 • Medium	(B) 35% and then 5%	Final prices: $2,000 \times 0.7 \times 0.9 = ₹1,260$; $2,000 \times 0.8 \times 0.8 = ₹1,280$; $2,000 \times 0.75 \times 0.85 = ₹1,275$; $2,000 \times 0.65 \times 0.95 = ₹1,235$. The lowest is with 35% and then 5%.
11	Part 3 • Ch3 • Medium	(A) It increases by 8%.	New area = $1.2 \times 0.9 = 1.08$ times the old area, so the area increases by 8%.
12	Part 4 • Ch4 • Easy	(D) $x^2 + 6x + 36$	$(x + 6)^2 = x^2 + 12x + 36$, not $x^2 + 6x + 36$. The others are $(x + 5)^2$, $(2x - 3)^2$ and $(3x + 4)^2$.
13	Part 4 • Ch4 • Easy	(C) 5	$x^2 - y^2 = (x + y)(x - y)$, so $45 = 9 \times (x - y)$ and $x - y = 5$.
14	Part 4 • Ch4 • Easy	(A) 180 cm	Let the shorter piece be x cm. Then $x + (2x + 40) = 250$, so $3x = 210$ and $x = 70$. The longer piece is $2 \times 70 + 40 = 180$ cm.
15	Part 4 • Ch4 • Medium	(B) 12	$k = a + b$ where $a \times b = 24$: the pairs 1×24 , 2×12 , 3×8 , 4×6 give $k = 25, 14, 11, 10$. So k cannot be 12.
16	Part 4 • Ch4 • Medium	(C) 80	Let there be x passengers. Then $\frac{2}{5}x + 12 = 44$, so $\frac{2}{5}x = 32$ and $x = 80$.
17	Part 5 • Ch5 • Easy	(D) 15 cm	$BD = 8$ cm. By the Baudhāyana–Pythagoras theorem, $h^2 = 17^2 - 8^2 = 289 - 64 = 225$, so $h = 15$ cm.

Q	Part • level	Answer	Why?
18	Part 5 • Ch5 • Easy	(A) One of its angles is 90°.	Adjacent angles of a parallelogram add up to 180°, so if one angle is 90°, all four are 90° and ABCD is a rectangle.
19	Part 5 • Ch5 • Easy	(B) a prism whose base has 10 sides	A prism with an n-sided base has $n + n + n = 3n$ edges, so $3 \times 10 = 30$. The others have 24, 20 and 45 edges.
20	Part 5 • Ch5 • Medium	(D) 18 cm	Perimeter of $\triangle ABC = 4 + 5 + 6 = 15$ cm, so the scale factor is $45 \div 15 = 3$. The longest side RP = $3 \times 6 = 18$ cm.
21	Part 5 • Ch5 • Medium	(C) 13 m	The difference in heights is $9 - 4 = 5$ m and the gap is 12 m. Distance = $\sqrt{(12^2 + 5^2)} = \sqrt{169} = 13$ m.
22	Part 6 • Ch6 • Easy	(B) Cone L	Volume = $\frac{1}{3}\pi r^2 h$, so compare $r^2 \times h$: K $36 \times 2 = 72$, L $25 \times 3 = 75$, M $4 \times 15 = 60$, N $9 \times 8 = 72$. Cone L has the greatest volume.
23	Part 6 • Ch6 • Easy	(C) 2	Volume of B = $\pi \times (2r)^2 \times (h \div 2) = \pi \times 4r^2 \times h \div 2 = 2\pi r^2 h$, which is 2 times the volume of A ($\pi r^2 h$).
24	Part 6 • Ch6 • Easy	(B) 23.1 cm ³	Radius = $4.2 \div 2 = 2.1$ cm and height = 50 mm = 5 cm. Volume = $\frac{1}{3} \times \frac{22}{7} \times 2.1 \times 2.1 \times 5 = 23.1$ cm ³ .
25	Part 6 • Ch6 • Medium	(A) 80	Volume = $60 \times 40 \times 40 = 96,000$ cm ³ = 96 litres. Number of jugs = $96 \div 1.2 = 80$.
26	Part 6 • Ch6 • Medium	(D) 9 cm	Cylinder = $\frac{22}{7} \times 3.5 \times 3.5 \times 10 = 385$ cm ³ , so the cone is $500.5 - 385 = 115.5$ cm ³ . Then $\frac{1}{3} \times 38.5 \times h = 115.5$, so $h = 9$ cm.
27	Part 7 • Ch8 • Easy	(A) Maths	An angle of 90° is one quarter of 360°, so the subject must have one quarter of 480 = 120 students: Maths ($120/480 \times 360^\circ = 90^\circ$).
28	Part 7 • Ch8 • Easy	(B) 10	Total = $18 + 8 + 14 + 6 + 4 + 10 = 60$ over 6 days. Mean = $60 \div 6 = 10$.
29	Part 7 • Ch8 • Medium	(C) (17.5, 10)	The frequency polygon joins the mid-points of the tops of the bars. The class 15–20 has mid-point 17.5 and frequency 10, so (17.5, 10) lies on it.
30	Part 8 • Ch9 • Easy	(C) Item C	Profit percents: A $20/80 = 25\%$, B $30/150 = 20\%$, C $15/50 = 30\%$, D $36/200 = 18\%$. Item C gives the greatest profit percent.
31	Part 8 • Ch9 • Easy	(A) ₹240	Interest for a year = $36,000 \times 8 \div 100 = ₹2,880$. For each month = $2,880 \div 12 = ₹240$.
32	Part 8 • Ch9 • Easy	(D) ₹1,380	15% of cost = ₹180, so cost = $180 \times 100 \div 15 = ₹1,200$. Selling price = $1,200 + 180 = ₹1,380$.
33	Part 8 • Ch9 • Medium	(B) 12.5%	Cost = $30 \times 60 + 20 \times 90 = ₹3,600$. Sale = $50 \times 81 = ₹4,050$. Profit = ₹450, and $450/3,600 \times 100 = 12.5\%$.
34	Part 8 • Ch9 • Medium	(D) 25%	If the selling price is 5 parts, the profit is 1 part and the cost is 4 parts. Profit percent = $1/4 \times 100 = 25\%$.
35	Part 9 • Ch10 • Easy	(A) 3 cm	Each step divides the side by 3: $81 \rightarrow 27 \rightarrow 9 \rightarrow 3$. At Step 3 each smallest square has side 3 cm.
36	Part 9 • Ch10 • Easy	(C) 5	The digit pairs are 1 and 8, 2 and 9, 0 and 7. They give 18, 81, 29, 92 and 70 (07 is not two-digit): 5 numbers.
37	Part 9 • Ch10 • Easy	(D) 48	The 4 colours repeat in full blocks all the way round, so the number of beads must be a multiple of 4. Only $48 = 4 \times 12$ is.
38	Part 9 • Ch10 • Medium	(B) 2,178	Four times the number still has 4 digits, so A is 1 or 2. A is the last digit of an even product, so A = 2. Then $4 \times D$ ends in 2 and $D \geq 8$, so D = 8. Checking gives $2,178 \times 4 = 8,712$.
39	Part 9 • Ch10 • Medium	(A) 18	Top shelf $1 \rightarrow (4, 5), (5, 4)$: 2 ways; top 2 $\rightarrow$ 3 ways; top 3 $\rightarrow$ 4 ways; top 4 $\rightarrow$ 5 ways; top 5 $\rightarrow$ 4 ways. Total = $2 + 3 + 4 + 5 + 4 = 18$.

Q	Part · level	Answer	Why?
40	Part 9 · Ch10 · Medium	(C) 5,832	Here $36 = 4 \times 9$, and 4 and 9 have no common factor. For 5,832, the last two digits 32 are divisible by 4, and $5 + 8 + 3 + 2 = 18$ is divisible by 9. Each of the others fails one test.
41	Section II · Ch4 + Ch10 · Hard	(D) 4	Let the number be $10a + b$. Then $10a + b = a + b + ab + 12$ gives $a(9 - b) = 12$. As $9 - b \leq 9$, $a \times (9 - b)$ can be 2×6 , 3×4 , 4×3 or 6×2 , giving 23, 35, 46 and 67 — 4 numbers.
42	Section II · Ch5 + Ch10 · Hard	(C) 8	Area 5 cm^2 needs a side $\sqrt{5} \text{ cm}$, which joins dots 1 across and 2 up ($1^2 + 2^2 = 5$). Such a square fills a $3 \text{ cm} \times 3 \text{ cm}$ block; there are 4 such blocks, and each holds 2 tilted squares: $4 \times 2 = 8$.
43	Section II · Ch4 + Ch5 · Hard	(A) 20 cm	Let $MF = x$ and $MD = y$. $\triangle LDE \sim \triangle LMN$ gives $(12 - y)/12 = x/6$, so $y = 12 - 2x$. Then $x(12 - 2x) = 16$ gives $x^2 - 6x + 8 = 0$, that is $(x - 2)(x - 4) = 0$. $x = 4$ makes a square, so $x = 2$, $y = 8$ and the perimeter is $2 \times (2 + 8) = 20 \text{ cm}$.
44	Section II · Ch3 + Ch9 · Hard	(B) 20%	Cost of a desk = $3,000 \times 0.75 \times 0.8 = ₹1,800$. Price to the school = ₹2,700. For 5 desks he spends $5 \times 1,800 = ₹9,000$ and receives $4 \times 2,700 = ₹10,800$. Profit = ₹1,800 = 20% of ₹9,000.
45	Section II · Ch6 + Ch9 · Hard	(C) 25%	Curved surface of one post = $2 \times 22/7 \times 7 \times 160 = 7,040 \text{ cm}^2$. For 25 posts: $176,000 \text{ cm}^2 = 17.6 \text{ m}^2$. Payment = $17.6 \times 50 = ₹880$. Profit = $880 - 704 = ₹176$, and $176/704 \times 100 = 25\%$.